

Fundamentals of Operating System Design

Introduction to Operating Systems

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Learning Outcomes

After completing this module, students will be able to:

- Define an Operating System.
- Explain why an Operating System is required.
- Describe the goals of an Operating System.
- Explain the evolution of Operating Systems.
- Identify the major responsibilities of an Operating System.

Why Study Operating Systems?

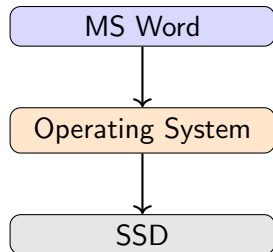
Think About It

What happens when you press

Save

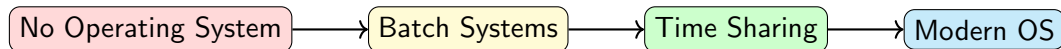
in Microsoft Word?

- Where is the file stored?
- Who communicates with the SSD?
- Who allocates memory?
- Who manages CPU time?
- Who protects your data?



Answer

Evolution of Computing

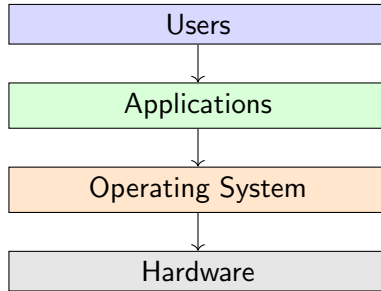


- Computers became faster.
- Users increased.
- Resources had to be shared.
- Operating Systems evolved to manage these resources efficiently.

Definition of Operating System

William Stallings

An Operating System is a program that controls the execution of application programs and acts as an interface between users and computer hardware.



Why Do We Need an Operating System?

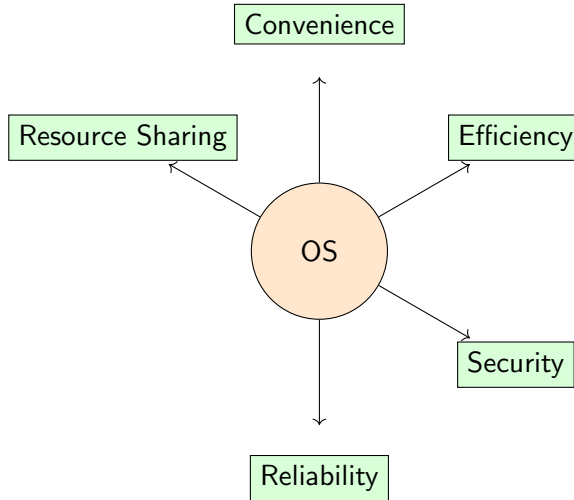
Without OS

- Difficult to use hardware
- No multitasking
- No security
- No file management
- No memory management

With OS

- Easy interaction
- Efficient CPU usage
- Memory allocation
- File organization
- Device management
- Protection and security

Goals of an Operating System



Summary

Key Points

- Operating System is system software.
- It acts as an interface between users and hardware.
- It evolved to efficiently manage computer resources.
- Main goals are
 - Convenience
 - Efficiency
 - Resource Sharing
 - Security
 - Reliability

Think

Can a modern computer work efficiently without an Operating System?